Walkthrough

Building an ESP32 Fencing Weapon Tester 2.0

1. Overview

This document describes the construction of an ESP32-based fencing weapon tester. It includes the complete bill of materials, PCB assembly, enclosure, power setup, firmware flashing, calibration, and final Wi-Fi/OTA configuration.

Reference repositories:

- https://github.com/pietwauters/esp32-scoring-device
- https://github.com/pietwauters/esp32_scoring_device_hardware
- https://github.com/pietwauters/remotecontrolapp
- https://github.com/pietwauters/WeaponWireTester

2. Bill of Materials (BOM)

The table below lists all electronic components required for the tester. The authoritative and up-to-date BOM can be found at:

https://github.com/pietwauters/esp32_scoring_device_hardware/blob/main/BOM.md

2.1 Electronic Components

Qty	Designator	Description	Value / Specs	Package / Size	Remarks
1*		PCB	Rev 1.7		Breadboard or perfboard possible
1	U1	ESP32 DevKit V1	30-pin (2×15)		Do NOT use 38-pin version
2	C1, C3	Electrolytic capacitor	470 µF – 35 V	Ø10 mm, 5 mm pitch	

2	C2, C4	Polyester film capacitor	0.1 µF	7.3 × 2.5 mm	
1	Q1	NPN transistor	BC337, 0.8 A 45 V	TO-92	
7	R1–R7	Metal film resistor	47 Ω, 1%	1/4 W	Tester resistors
1	R8	Metal film resistor	3.3 kΩ, 1%	1/4 W	
1	R12	Metal film resistor	10 kΩ, 1%	1/4 W	
6		Female banana jack	4 mm	Panel mount	
1		Female banana jack	3 mm	Panel mount	
1*	J9	JST connector	1 × 2	2.54 mm pitch	Populate as required
1		Li-Ion battery	1000 mAh	102050	
1		LED matrix panel	WS2812B (25 LEDs)	38 × 38 mm	
1		Slide switch mounting on board		7 mm	
1		USB-C Li-Ion charger board	LX-LCBST		

1		0.91 Inch 6Pin IIC I2C SPI Interface OLED White LCD Display Module
1		Right Angle 90 degree Single Row Male PCB Board Connector

3. Enclosure

Download the latest STL files for the enclosure and inserts from:

https://www.printables.com/model/1006755-body-wire-weapon-test-box

4. Mechanical Hardware

- 4 × M3 × 14 screws (cylinder or round head)
- 4 × M3 hot-melt threaded inserts
- 4 × M2 × 4 screws (PCB mounting)
- 4 × M2 hot-melt threaded inserts
- Silicone cable, 24 AWG
- 1 × M5 × 12 bolt (DIN 933)
- 1 × M5 autolock nut
- 1 × M5 acorn nut
- 7 x 6 mm 0.75 mm thread nuts

5. PCB Assembly

Assemble the PCB except for the screens.

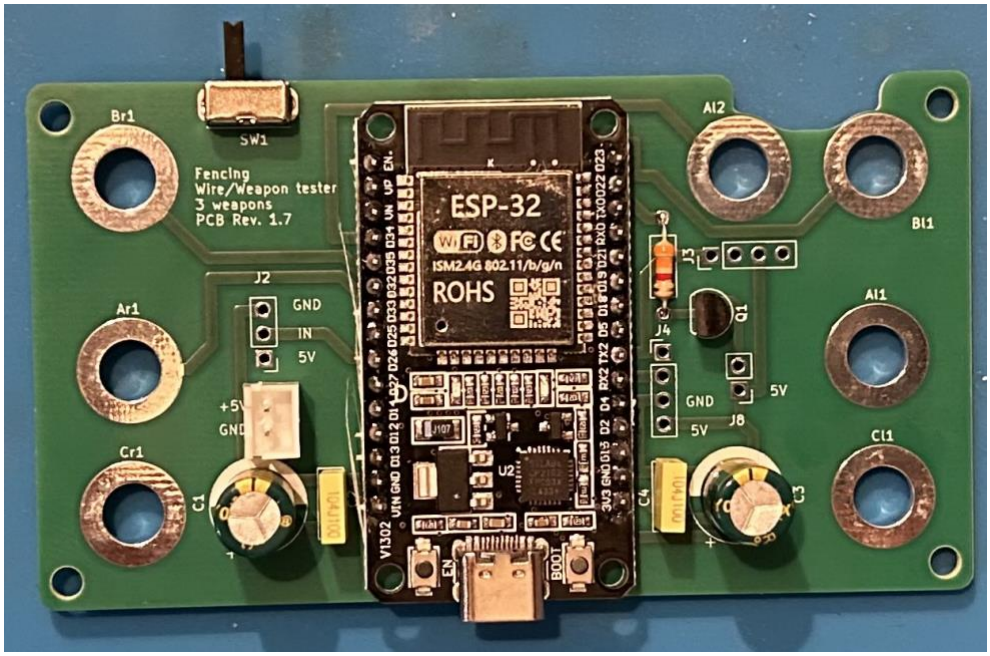
IMPORTANT: Connector J9 polarity may be reversed depending on the power input. Always verify the polarity of the J2 male connector before powering the board.

IMPORTANT: Program the ESP32 (see below) before soldering it in order to prevent installing a defective one which could not be replaced.

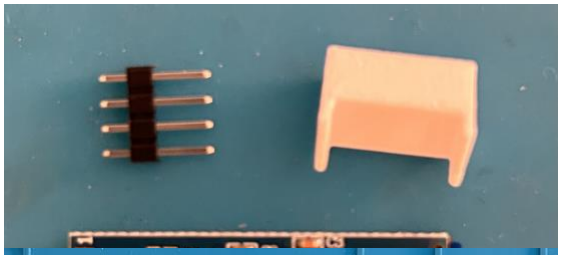
To do so install the correct USB driver depending on your ESP32 board (CH340C or CP210x).

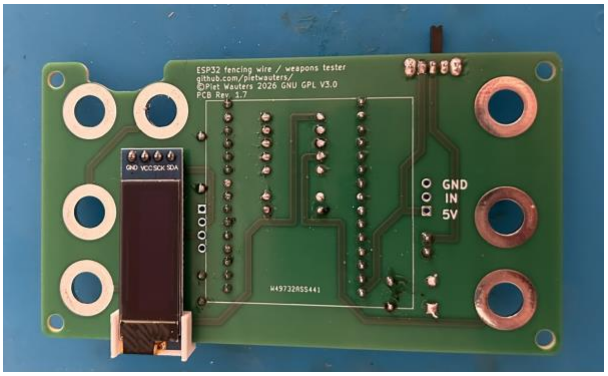
Flash the ESP32 using: https://espressif.github.io/esp-launchpad/

Firmware file: ESP_FIRMWARE_TESTER.

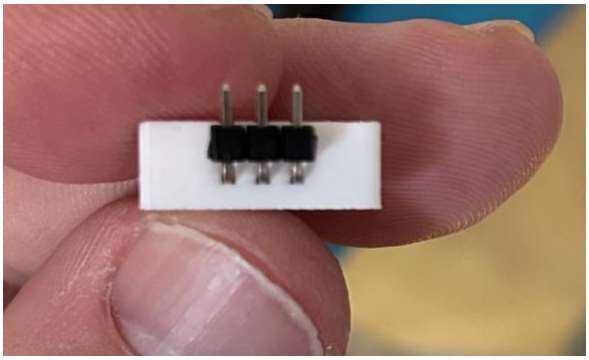


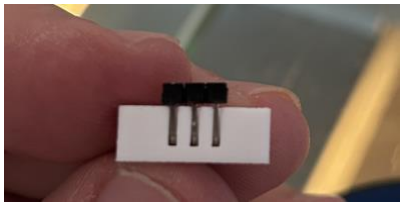
Mount the OLED and the 5x5 LED Panel

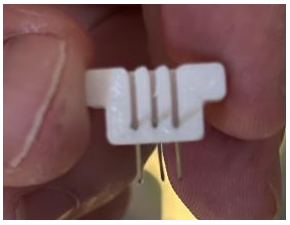


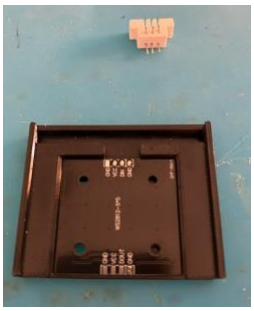


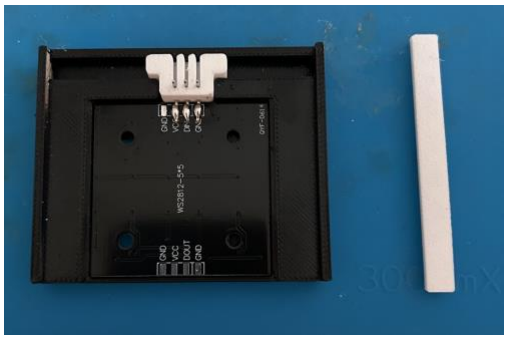


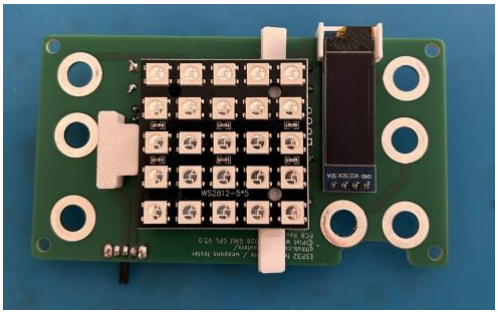




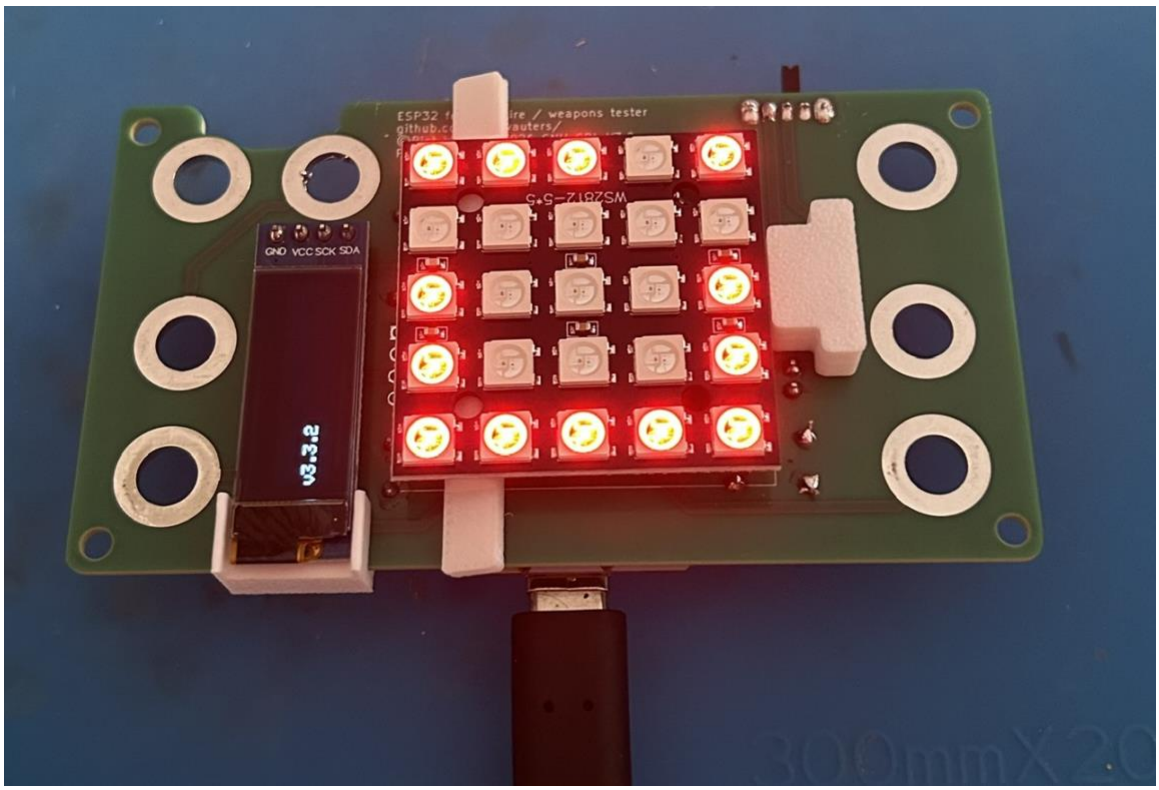








Plug the ESP to a 5V power source to check that the displays work properly.

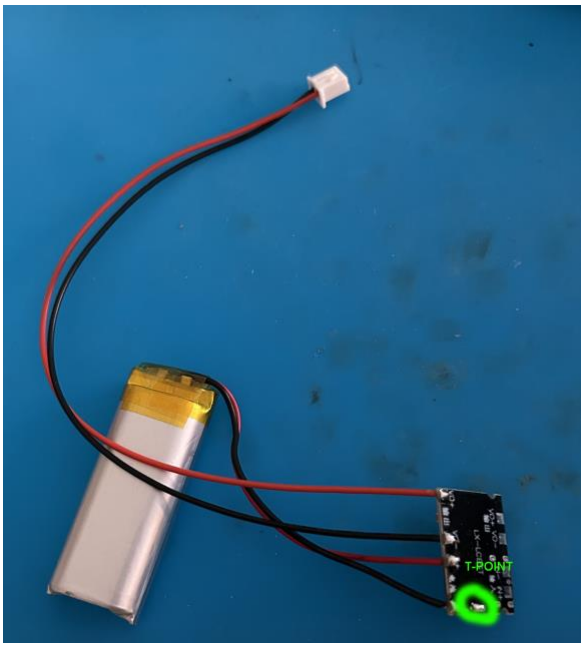


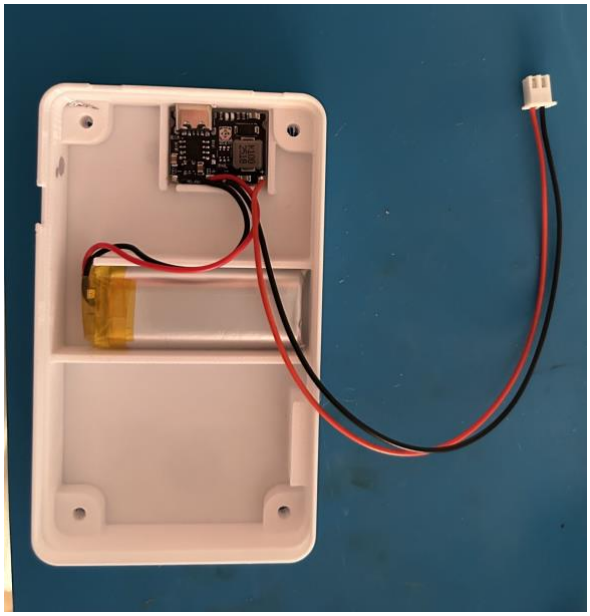
6. Power Setup

6.1 Battery Installation

Connect the battery to the charging circuit. Close the T-point (green)to enable cold boot,

Place the battery inside the enclosure (place double side tape below) and place the charging circuit in the enclosure with hot glue.





6.2 Output Voltage Adjustment

Adjust the potentiometer on the charging board until the output voltage reaches exactly 5.0 V.

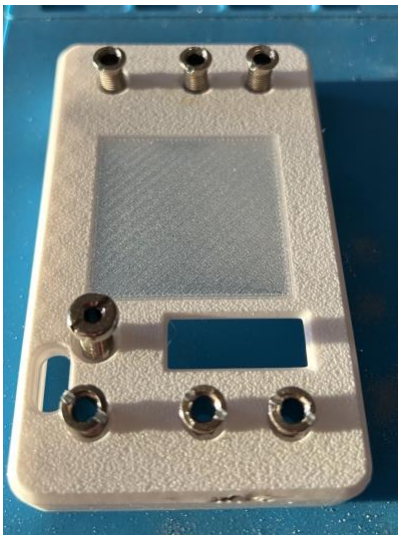
6.4 Power Test

Connect the J2 connector and positive and negative outputs from the charging circuit and switch to the PCB. Verify that the board powers up correctly.

7. Mounting the cover

Get 6x4 mm and 1x3 mm female banana jack with their nuts. Mount them on the upper part of the case. If your jack have no groove Dremel it. Thight them with a Screwdriver.





Place the insets and then the board with the switch cover. Tight all nuts except Cl1



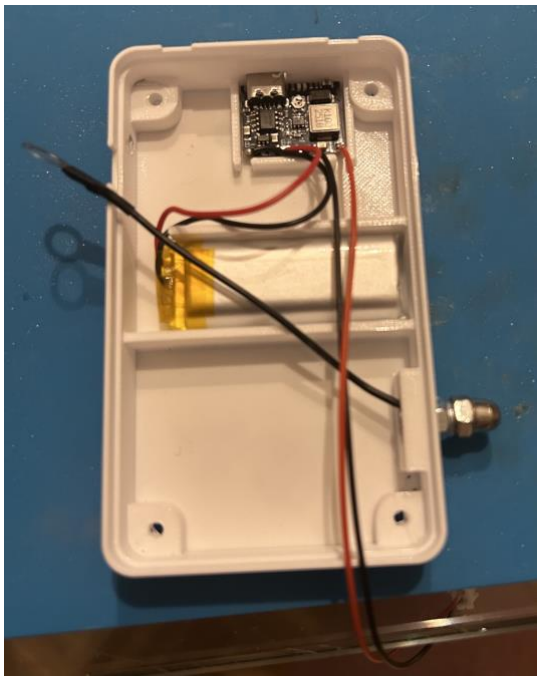


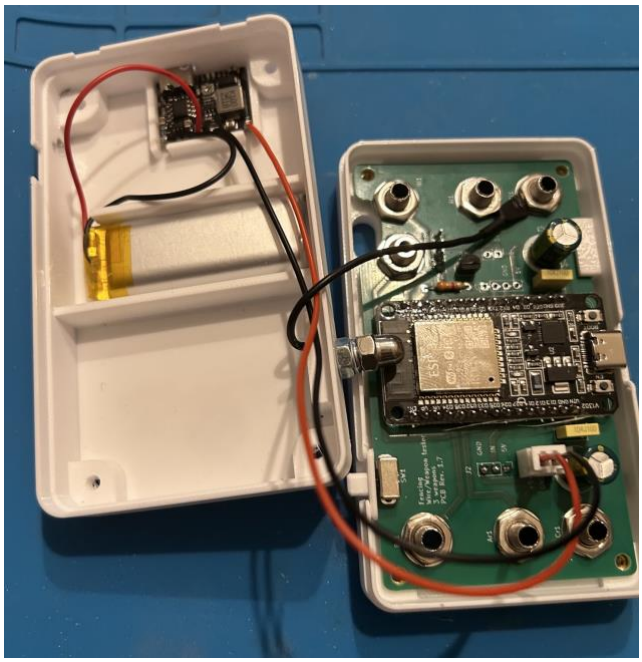


Get the 1 × M5 × 12 bolt (DIN 933), 1 × M5 autolock nut and 1 × M5 acorn nut, prepare a cable with a 6 mm terminal to connect the bolt to the board.



Place the screw locking the cable in the lower part of the case and close it with both nuts.





Now you are ready for calibration.

8. Calibration

Prepare a set of known resistors ranging from 1 to 10 Ω, a 33 one and crocodile clips.

Connect the ESP32 via USB-C and use https://webserial.app/.

Follow the on-screen instructions and connect the resistors to the C sockets as requested.

If calibration fails, erase the flash using the ESP Launchpad tool and reflash the firmware.

When calibrated close the box with the 4x25 mm M3 screws.

9. Wi-Fi and OTA Configuration

After power-up, the ESP32 creates a Wi-Fi access point:

SSID: TESTER

Password: 01041967

Interfaces:

OTA updates: http://192.168.4.1/update

Settings: http://192.168.4.1/settings